

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

FACULTY OF TECHNOLOGY & ENGINEERING

SMT. KUNDANBEN DINSHA PATEL DEPARTMENT OF INFORMATION TECHNOLOGY

Subject Name: Design and Analysis of Algorithm
Subject Code: IT341

Semester: B.Tech V
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Practical List

Analysis of Program should contain following sub heading(s).

1. **Impact of Input Size on the Performance of Program.** Make Table and Draw graph of Input Size Vs Running Time/Total No of Instructions. Take at least Five Input of Different Size.
2. **Impact of Input Quality on the Performance of Program.** Make Table and Draw graph of Best Case, Worst Case and Average Case Input Quality Vs Running Time/ Total No. of Instructions.
3. **Rate of Growth of Program.** Make Table and Draw Graph of Input Size Vs Instruction(s) Running Maximum No of Time in the Program.
4. **Conclusion** from the above graph or Data Table.

Sr No.	Practical Aim	Hrs
1	Implement and analyze algorithms given below.	02
1.1	Fibonacci Series(Iterative and Recursive)	
1.2	Factorial of a given number (Iterative and Recursive)	
2	Analyze algorithms given below.	04
2.1	Bubble Sort	
2.2	Selection Sort	
2.3	Insertion Sort	
3	Implement and analyze algorithms given below.(Divide and Conquer Strategy)	04
3.1	Design and implement searching algorithm to find given word from English dictionary using minimum number of comparisons. Also find out time complexity of algorithm.	
3.2	Implement and compare worst case performance of Merge Sort and Quick Sort.	
4	Implement and analyze below given problem.(Greedy Approach)	06
4.1	A cashier at any mall needs to give change of an amount to customers many times in a day. Cashier has multiple number of coins available with different denominations which is described by a set C. Implement the program for a cashier to find the minimum number of coins required to find a change of a particular amount A. Output should be	

the total number of coins required of given denomination.

Check the program for following test cases:

Test Case	Coin denominations C	Amount A
1	₹1, ₹2, ₹3	₹ 5
2	₹18, ₹17, ₹5, ₹1	₹ 22
3	₹100, ₹25, ₹10, ₹5, ₹1	₹ 289

Is the output of Test case 2 is optimal? Write your observation.

- 4.2 A Burglar has just broken into the Fort! He sees himself in a room with n piles of gold dust. Because the each pile has a different purity, each pile also has a different value ($v[i]$) and a different weight ($w[i]$). A Burglar has a bag that can only hold W kilograms.

Given $n=5$, $v=\{4,2,2,1,10\}$, $c=\{12,1,2,1,4\}$ and $W=15$, calculate which piles Burglar should completely put into his bag and which he should put only fraction into his bag.

Design and implement an algorithm to get maximum piles of gold using given bag with W capacity, Burglar is also allowed to take fractional of pile.

- 4.3 There are eleven Professors in a Department. Each professor wants to deliver lecture in same day. Each professor has some time limits for lecture. Professor earns credit if and only if lecture is arranged on or before its time limit.

Professor	1	2	3	4	5	6	7	8	9	10	11
credit	78	90	50	60	75	10	80	55	88	74	59
Lecture Limit	5	4	5	3	2	1	4	6	4	5	6

Design and implement greedy approach to schedule maximum number of lectures in the department without and with credit.

5 Implement and analyze given problems (Dynamic Programming)

06

- 5.1 Given two integer arrays $val[0..n-1]$ and $wt[0..n-1]$ which represent values and weights associated with n items respectively. Also given an integer W which represents knapsack capacity, find out the maximum value subset of $val[]$ such that sum of the weights of this subset is smaller than or equal to W . You cannot break an item, either pick the complete item, or don't pick it (0-1 property).

- 5.2 Given a chain $\langle A_1, A_2, \dots, A_n \rangle$ of n matrices, where for $i=1,2,\dots,n$ matrix A_i with dimensions. Implement the program to fully parenthesize the product $A_1, A_2, \dots, A_n$ in a way that minimizes the number of scalar multiplications.

Also calculate the number of scalar multiplications for all possible combinations of matrices.

Test Case	n	Matrices with dimensions
1	3	$A_1: 3*5, A_2: 5*6, A_3: 6*4$
2	6	$A_1: 30*35, A_2: 35*15, A_3: 15*5, A_4: 5*10, A_5: 10*20, A_6: 20*25$

- 5.3 Implement a program which has BNMCOEF() function that takes two parameters n and k and returns the value of Binomial Coefficient $C(n, k)$. Compare the dynamic programming implementation with recursive implementation of BNMCOEF(). (In output, entire table should be displayed.)

Test Case	n	k
1	5	2
2	11	6
3	12	5

- 5.4 Implement a program to print the longest common subsequence for the following strings:

Test Case	String1	String2
1	ABCDAB	BDCABA
2	EXPONENTIAL	POLYNOMIAL
3	LOGARITHM	ALGORITHM

6 String Matching Problem.

02

- 6.1 Suppose you are given a source string $S[0..n-1]$ of length n, consisting of symbols a and b. Suppose further that you are given a pattern string $P[0..m-1]$ of length $m < n$, consisting of symbols a, b, and *, representing a pattern to be found in string S. The symbol * is a “wild card” symbol, which matches a single symbol, either a or b. The other symbols must match exactly. The problem is to output a sorted list M of valid “match positions”, which are positions j in S such that pattern P matches the substring $S[j..j+|P|-1]$. For example, if $S = ababbab$ and $P = ab*$, then the output M should be [0, 2].

Implement Naive and Rabin karp algorithm to solve the problem.

7 Implement and analyze the problem.

02

- 7.1 Eight Queen Problem

8 Design and analyze below problems (Graph)

04

- 8.1 Given an undirected graph and a number m, determine if the graph can be colored with at most m colors such that no two adjacent vertices of the graph are colored with same color. Here coloring of a graph means assignment of colors to all vertices. Solve this using Backtracking.
- 8.2 Write a program to detect cycles in directed graph.
- 8.3 Let us consider the situation of a war. In your country, there is a network of telephone lines between 9 cities (A, B, C...H, I) i.e. the 9 cities are connected by telephone line, which means that a message from one city to any other city can be transmitted through the line. Like we can transfer message from city A to city B even though they are not "directly" connected by a line. So what's the catch, everything seems fine, right?

You are the "army-general" of your country and you've to take a decision, you have to find the city which, if damaged would incur the greatest network blockage (Considering that damaging the city damages all the connected telephone lines in it).

Implement this situation and find out Which city would you try to protect the most and why?

